

Tutorial 2 Solutions
SCR forced commutation circuits

1.

Resonant pulse commutation circuit / class B commutation

$$I_0 = 300A$$

Peak value of resonant current

$$I_p = V_s \sqrt{\frac{C}{L}} = 230 \sqrt{\frac{20}{6}} = 460A$$

Resonant frequency

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{10^6}{\sqrt{100}} = 0.1 \times 10^6 \text{ rad/s}$$

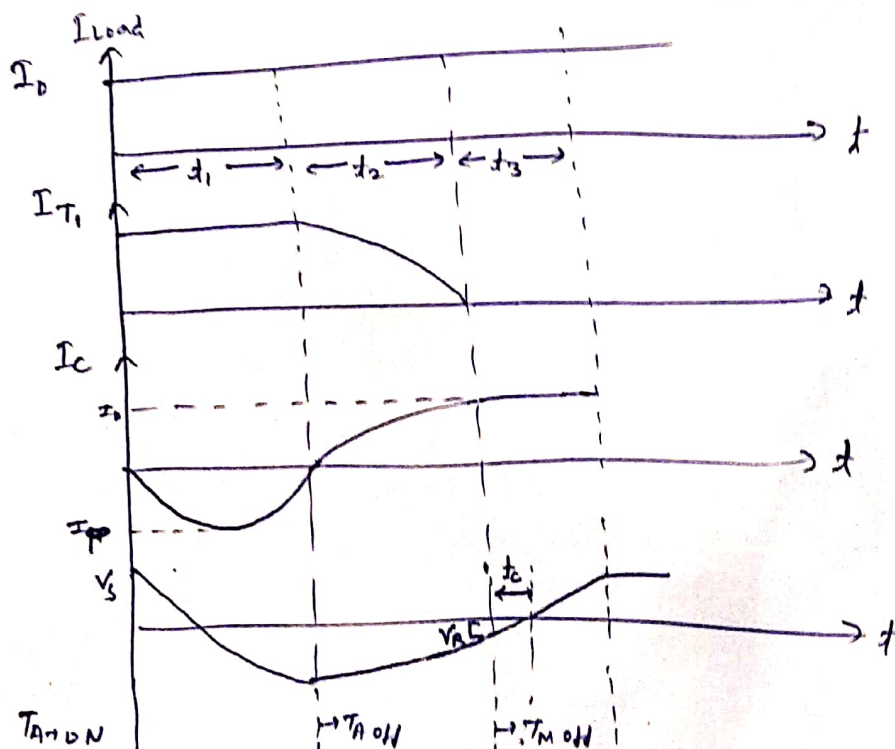
(a) Conduction time for Auxiliary thyristor

$$= \frac{\pi}{\omega_0} = \frac{\pi}{0.1 \times 10^6} = 31.416 \mu s$$

(b)

AS we know

$$i_c(t) = -V_s \sqrt{\frac{C}{L}} \sin \omega_0 t$$



Voltage across capacitor at the instant when main thyristor is off will be $(-V_R)$

we know

$$V_c(t) = V_s \cos \omega_0 t$$

$$V_s \cos(\pi + \omega_0 t_2) = -V_R$$

$$I_p \sin \omega_0 t_2 = I_0$$

$$\omega t_2 = \sin^{-1}\left(\frac{I_0}{I_p}\right)$$

voltage across thyristor

$$= V_s \cos\left(\pi + \sin^{-1}\left(\frac{I_0}{I_p}\right)\right)$$

$$= -V_s \cos\left(\sin^{-1}\left(\frac{I_0}{I_p}\right)\right)$$

$$= -230 \cos\left(\sin^{-1}\left(\frac{300}{460}\right)\right)$$

$$= -174.35 \text{ V}$$

(c) circuit turn off time of main thyristor:
i.e. Time for which T_1 is in Reverse biased

when capacitor voltage is

$(-V_R \text{ to } 0) \rightarrow$ will be t_c

During this period capacitor current is I_0

So,

$$t_c = \frac{C V_R}{I_0} = 20 \times 10^{-6} \times \frac{174.355}{300}$$

$$= 11.624 \mu\text{s}$$

2.

Load current

